Federated Learning Approach for Obstructive Sleep Apnea Prediction

Bachelor thesis presentation

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Outline

Introduction

- Team
  - **Manja Gorenc Novak**, *Head of AI and Senior Data Scientist*
  - Rok Vinder, *Data Analyst and Machine Learning Engineer*
- Result d.o.o
  - Digital solutions
- Context
  - eEarly
    - Health data
    - Obstructive sleep apnea

Project background

Problem statement and proposed solution

- Model consists of weights, imagine “knowledge”
- Problem statement
  - Standard machine learning
  - Data to the model (centralized)
  - Private (health) data
  - Regulations
  - Bandwidth
- Proposed solution
  - Proof-of-concept
  - Federated machine learning
  - Private data is stationary
  - Model to the data (decentralized)

Project background

Stakeholders and their needs

- **Developers**
  - Understand code
  - Interpret results
  - Run the project
- *Patients*
  - **Data privacy**
- *Doctors*
  - *Improved diagnosis*
  - *Better tools*

Research and Development

Comparative study and technology decision

Flower

- beginner-friendly
- scalable

NVIDIA FLARE

- enterprise ready
- heavy for small projects

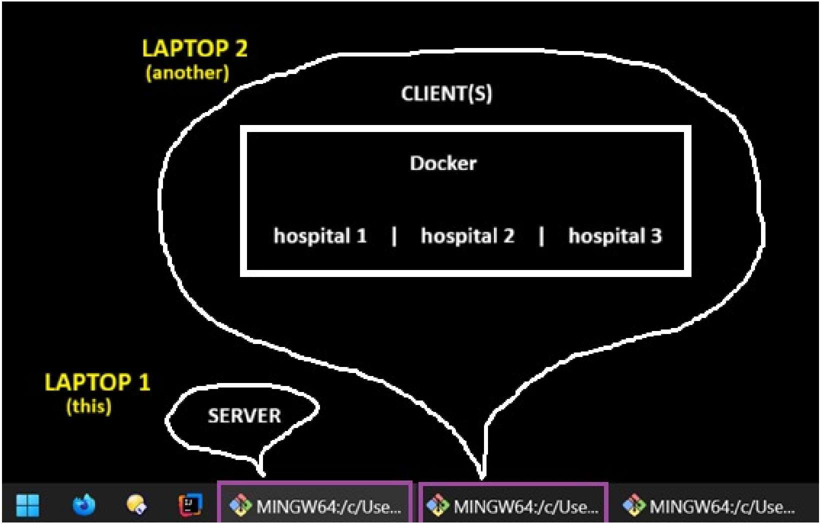
TensorFlow Federated

- complex
- the least integration

Research and Development

Federated learning algorithm

Demo


LAPTOP 2
(another)
CLIENT(S)
Docker
hospital 1 | hospital 2 | hospital 3
LAPTOP 1
(this)
SERVER
MINGW64:/c/Use...
MINGW64:/c/Use...
MINGW64:/c/Use...

Reflection

Improvements, and choice reflection

Improvements and weaknesses

- Evaluation metrics
- Separate networks
- Real TLS certification

Choice reflection

- Excellent
- No change

Reflection

Achieved objectives

- Larger dataset
- Approach comparison
- Metric difference ~ 0.03 (train)

Importance	**Requirement**	**Done**
Must have	Single-machine and non-containerized simulation of apnea predictions using federated learning.	Yes
Should have	Single-machine and containerized (with Docker) $-\|\|-$	Yes
Could have	**Multi-machine and containerized (with Docker)** $-\|\|-$	Yes
Could have	Multi-machine and containerized (on Kubernetes) $-\|\|-$	No
Could have	Differential privacy mechanism implemented.	No

Table: Red color represents MVP, orange and blue represent broader or ultimate scope. Bold requirements was shown in the demo.

Reflection

Added value and impact

- Privacy (patients)
- Regulations (project managers, lawyers, developer)
- Modularity, readability, and scalability (developer)
- Project opportunities (Result)
- Better diagnosis tools (doctors)
- Less network usage (climate)
- Research community (scientists)
  - Data scarcity
  - Vision

Project Approach

Development method and collaboration

Scrumban

- Agile mindset
- Daily meetings, backlog
- Typical kanban board

Collaboaration

- Daily meetings, Slack, and Outlook

Acknowledgements

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Result d.o.o

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Thank you for your attention!

Questions and Answers